

SHETH LUJ AND SIR MV COLLEGE
Subject: Data Analysis with SAS / SPSS / R

M2 Practical No: 10

Aim: Creating graphical reports using ,ggplot2 (R).

Code:

```
library(ggplot2)
library(dplyr)
library(tidyr)
library(readr)
```

```
cancer_data <- read_csv("Cancer.dataset.csv", n_max = 20)
```

```
cancer_long <- cancer_data %>%
  select(id, radius_mean, texture_mean) %>%
  pivot_longer(cols = c("radius_mean", "texture_mean"),
               names_to = "Metric",
               values_to = "Value")
```

```
plot1 <- ggplot(cancer_long, aes(x = factor(id), y = Value, color = Metric, group = Metric)) +
  geom_line(size = 1) +
  geom_point(size = 2) +
  theme_minimal() +
  scale_color_manual(values = c("radius_mean" = "#2ECC71", "texture_mean" = "#9B59B6"),
                    labels = c("Mean Radius", "Mean Texture")) +
  labs(title = "Comparative Analysis: First 20 Patients",
       x = "Patient ID", y = "Value") +
  theme(axis.text.x = element_text(angle = 45, hjust = 1), legend.position = "bottom")
```

```
plot2 <- ggplot(cancer_long, aes(x = factor(id), y = Value, fill = Metric)) +
  geom_col() +
  facet_wrap(~Metric) +
  theme_minimal() +
  scale_fill_manual(values = c("radius_mean" = "#5DADE2", "texture_mean" = "#E74C3C")) +
  labs(title = "Clinical Metrics by Patient ID") +
  theme(axis.text.x = element_text(angle = 45, hjust = 1), legend.position = "none")
```

```
diagnosis_summary <- cancer_data %>% count(diagnosis)
plot3 <- ggplot(diagnosis_summary, aes(x = "", y = n, fill = diagnosis)) +
  geom_bar(stat = "identity", width = 1) +
  coord_polar("y", start = 0) +
  theme_void() +
```

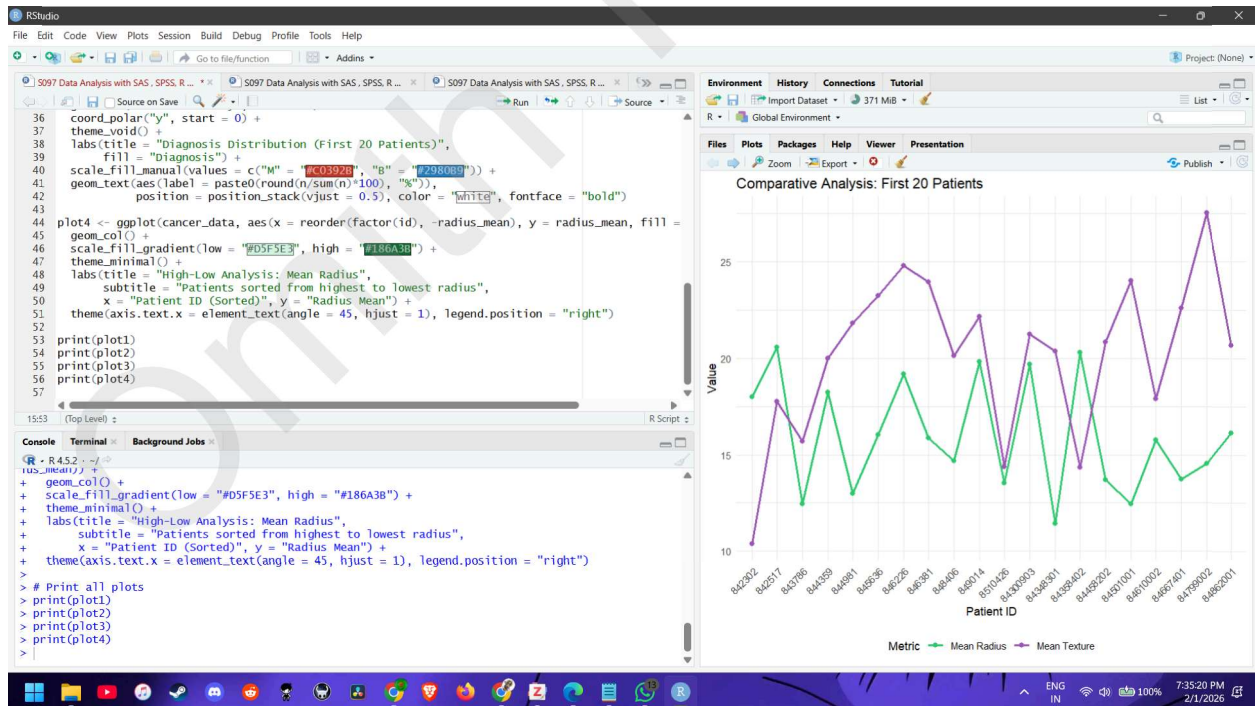
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```
labs(title = "Diagnosis Distribution (First 20 Patients)",  
      fill = "Diagnosis") +  
scale_fill_manual(values = c("M" = "#C0392B", "B" = "#2980B9")) +  
geom_text(aes(label = paste0(round(n/sum(n)*100), "%")),  
          position = position_stack(vjust = 0.5), color = "white", fontface = "bold")
```

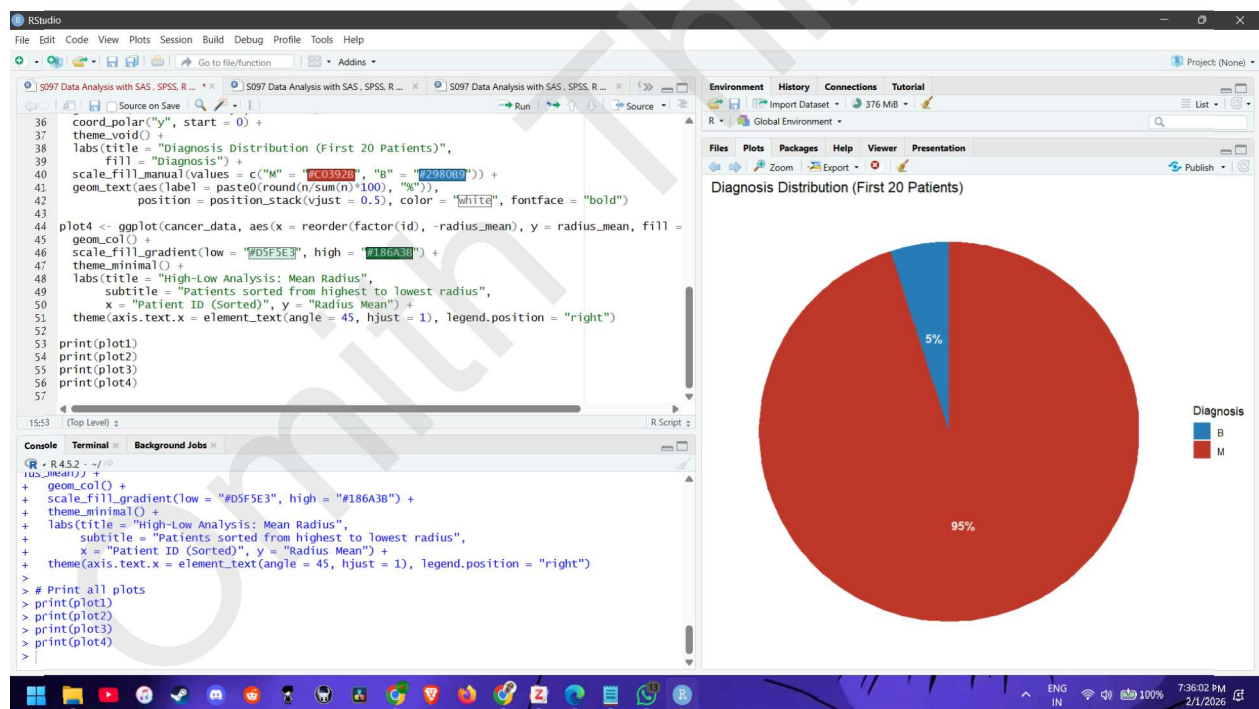
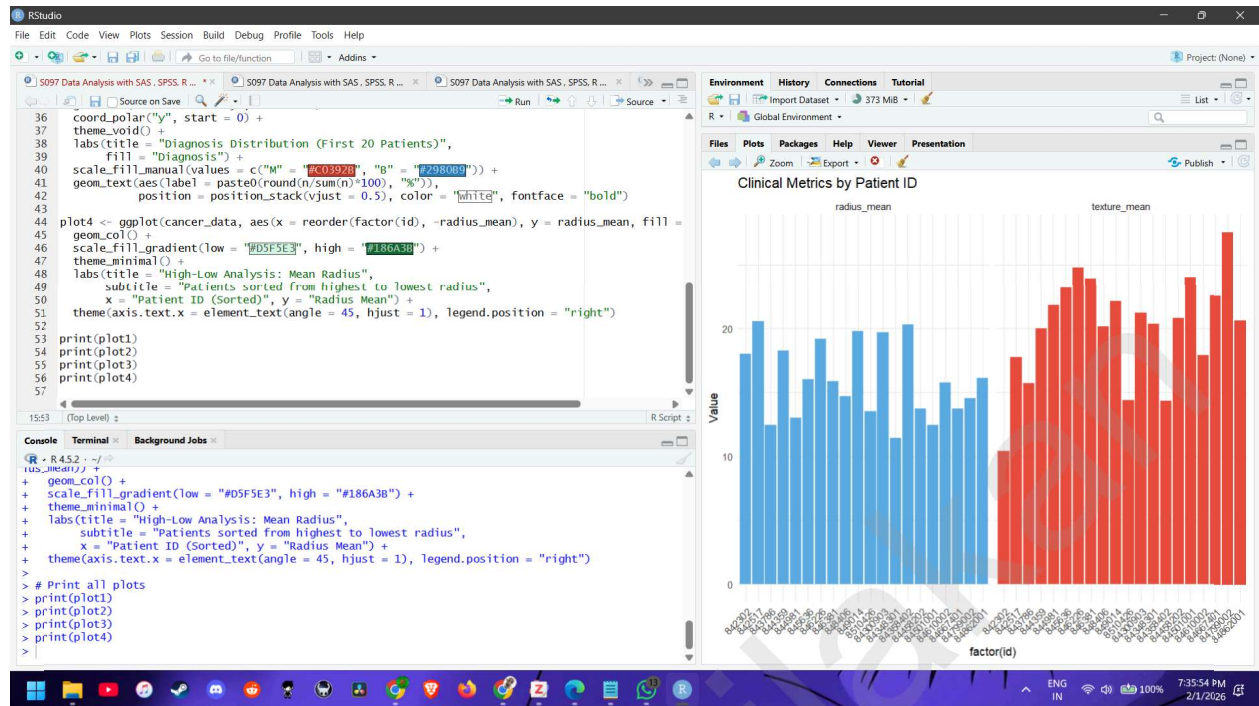
```
plot4 <- ggplot(cancer_data, aes(x = reorder(factor(id), -radius_mean), y = radius_mean, fill =  
radius_mean)) +  
  geom_col() +  
  scale_fill_gradient(low = "#D5F5E3", high = "#186A3B") +  
  theme_minimal() +  
  labs(title = "High-Low Analysis: Mean Radius",  
        subtitle = "Patients sorted from highest to lowest radius",  
        x = "Patient ID (Sorted)", y = "Radius Mean") +  
  theme(axis.text.x = element_text(angle = 45, hjust = 1), legend.position = "right")
```

```
print(plot1)  
print(plot2)  
print(plot3)  
print(plot4)
```

Output:



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